General Binary Valuations (Goods and Chores)

Scratch draft — not for circulation

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What this document is. A scratch working draft for the general-binary extension (every marginal in $\{-1, 0, 1\}$: each item may be a good for some agents and a chore for others), independent of `main.tex` and `working.tex`. Nothing here is promoted into either of those documents without an explicit decision once a result exists. See `docs/PS2_general_binary.md` for the problem statement and `docs/RESIDUAL_GENERAL_BINARY.md` for the running research log.

1 The Model and the Conjecture

1.1 General binary valuations

Let $N = \{1, \ldots, n\}$ be a set of agents and M a finite set of m indivisible items. Each agent $i \in N$ has a valuation $v_i : 2^M \to \mathbb{R}$ with $v_i(\emptyset) = 0$.

Definition 1 (General binary valuation)**.** *A valuation v_i is* general binary *if every marginal lies in $\{-1, 0, 1\}$:*

$$v_i(S \cup \{g\}) - v_i(S) \;\in\; \{-1, 0, 1\} \qquad \text{for every } S \subseteq M \text{ and every } g \in M \setminus S.$$

Nothing further is assumed: v_i need not be additive, submodular, subadditive, or monotone, and no relation is assumed between the valuations of different agents. Two consequences of the definition are worth stating explicitly, since both are excluded in the special cases this class generalizes.

- A single item may be a *good* for one agent and a *chore* for another: $v_i(g \mid S) = 1$ while $v_j(g \mid T) = -1$.
- For a *single* agent, an item's marginal may be positive on one set and negative on another: $v_i(g \mid S) = 1$ and $v_i(g \mid T) = -1$ for $S \neq T$. Such an item is neither a good nor a chore for that agent (Lemma 10).

We write $v_i(g \mid S) := v_i(S \cup \{g\}) - v_i(S)$ for a marginal.

An *allocation* $\mathcal{A} = (A_1, \ldots, A_n)$ is a tuple of pairwise disjoint subsets of M, assigned one per agent; it is *complete* if $\bigcup_i A_i = M$. A subsidy vector is $p \in \mathbb{R}^n_+$, and agents have quasi-linear utilities.

Definition 2 (Envy-free solution)**.** *An allocation $\mathcal{A}$ and a subsidy vector p constitute an* envy-free solution *$(\mathcal{A}, p)$ if*

$$v_i(A_i) + p_i \;\geq\; v_i(A_j) + p_j \qquad \text{for all } i, j \in N.$$

An allocation is envy-freeable *if some $p \in \mathbb{R}^n_+$ makes $(\mathcal{A}, p)$ an envy-free solution.*

For an allocation $\mathcal{A}$, the *envy graph $G_{\mathcal{A}}$* is the complete weighted digraph on N in which the arc (i, j) carries weight

$$w_{\mathcal{A}}(i, j) \;:=\; v_i(A_j) - v_i(A_i), \tag{1}$$

and $\ell_{\mathcal{A}}(i)$ denotes the maximum weight of a directed path in $G_{\mathcal{A}}$ starting at i, the empty path of weight 0 included.

1.2 The conjecture

Conjecture 3. *For every instance $\langle N, M, \{v_i\}_{i \in N}\rangle$ in which every v_i is general binary, there exist a complete allocation $\mathcal{A}$ and a subsidy vector $p \in \{0, 1\}^n$, with $\sum_{i \in N} p_i \leq n - 1$, such that $(\mathcal{A}, p)$ is an envy-free solution. Moreover $\mathcal{A}$ and p can be computed in polynomial time given value-oracle access to the v_i.*

Conjecture 3 is the common generalization of two theorems.

Observation 4 (The two boundary cases are known)**.** *Restricting Definition 1 to marginals in $\{0, 1\}$ gives the* dichotomous *class, for which Conjecture 3 is Theorem 4 of Barman, Krishna, Narahari and Sadhukhan [BKNS22]. Restricting it to marginals in $\{-1, 0\}$ gives the* negative dichotomous *class, for which Conjecture 3 is the main theorem of the companion document* `report/main.tex`*, proved for every n.*

Both boundary results are stated at the same generality as Definition 1 within their own sign restriction — neither assumes additivity, submodularity or subadditivity — so Conjecture 3 adds exactly one thing to what is known: the two signs may occur in the same instance.

1.3 What transfers

The following facts hold for general binary valuations and may be used without re-derivation.

Observation 5 (Integrality)**.** *Let v_i be general binary. Then $v_i(S) \in \mathbb{Z}$ and $|v_i(S)| \leq |S|$ for every $S \subseteq M$; every arc weight $w_{\mathcal{A}}(i, j)$ is an integer; and for every envy-freeable allocation $\mathcal{A}$ the minimal subsidy vector satisfies $p^* \in \mathbb{Z}^n_{\geq 0}$.*

Proof Fix i and enumerate $S = \{g_1, \ldots, g_k\}$. Telescoping from $v_i(\emptyset) = 0$ writes $v_i(S) = \sum_{u=1}^{k} v_i\big(g_u \mid \{g_1, \ldots, g_{u-1}\}\big)$, a sum of k terms each in $\{-1, 0, 1\}$ by Definition 1; hence $v_i(S) \in \mathbb{Z}$ and $|v_i(S)| \leq k$. Each arc weight (1) is a difference of integers, so the weight of any directed path is an integer. By Theorem 7, $p^*_i = \ell_{\mathcal{A}}(i)$ is a maximum of such weights together with 0, hence a non-negative integer. □

Observation 5 is what makes Conjecture 3 a statement about a *bound* rather than about rounding: once p^* is known to be integral, proving $p^*_i \leq 1$ for every i is the same as proving $p^* \in \{0,1\}^n$.

The two structural results of Halpern and Shah [HS19] transfer verbatim. Their proofs use only the arc weights (1) and permutations of the bundles; no additivity, no monotonicity and no sign condition on v_i enters, which is also the generality at which [BKNS22] invokes them.

Theorem 6 ([HS19])**.** *For any allocation $\mathcal{A} = (A_1, \dots, A_n)$ the following are equivalent.*

(i) $\mathcal{A}$ is envy-freeable.

(ii) $\mathcal{A}$ maximises utilitarian welfare across all reassignments of its own bundles: $\sum_i v_i(A_i) \geq \sum_i v_i(A_{\sigma(i)})$ for every permutation σ of N.

(iii) $G_{\mathcal{A}}$ has no positive-weight directed cycle.

Theorem 7 ([HS19])**.** *For any envy-freeable allocation $\mathcal{A}$, the subsidy vector p^* given by $p^*_i := \ell_{\mathcal{A}}(i)$ realises an envy-free solution $(\mathcal{A}, p^*)$, and $p^*_i \leq p_i$ for every i and every p with $(\mathcal{A}, p)$ an envy-free solution.*

Proposition 8 (Some agent is unpaid)**.** *For every envy-freeable allocation $\mathcal{A}$ there is an agent h with $p^*_h = 0$. Consequently, if $p^* \in \{0,1\}^n$ then $\sum_{i \in N} p^*_i \leq n-1$.*

Proof By Theorem 6(iii) the graph $G_{\mathcal{A}}$ has no positive-weight cycle, so a walk of maximum weight may be taken simple and $\ell_{\mathcal{A}}$ is finite; it is non-negative because the empty path qualifies. Choose a path P attaining $\max_i \ell_{\mathcal{A}}(i)$ and let h be its final vertex. If $\ell_{\mathcal{A}}(h) > 0$ there is a positive-weight walk Q leaving h, and P followed by Q is a walk from P's start of weight $w_{\mathcal{A}}(P) + w_{\mathcal{A}}(Q) > w_{\mathcal{A}}(P)$, contradicting the maximality of P among walks. Hence $\ell_{\mathcal{A}}(h) = 0$. The second claim follows because at most $n-1$ coordinates of p^* can then equal 1. $\square$

Proposition 8 means the bound on the total in Conjecture 3 is not an independent requirement but a consequence of the per-agent one, so the conjecture is the single-clause statement

$$\exists \mathcal{A} : \quad \max_{i \in N} \ell_{\mathcal{A}}(i) \leq 1.$$

1.4 Tightness

Example 9 (The $n-1$ lower bound)**.** Take n agents and a single item g with $v_i(\{g\}) = 1$ for every i. Every v_i is general binary. Under any complete allocation one agent holds g and the other $n-1$ hold nothing; each of those $n-1$ agents envies the holder by exactly 1, so every envy-free solution assigns each of them a subsidy at least 1 more than the holder's. The minimum total is therefore $n-1$.

The same bound is forced from the opposite sign: n agents and $m = n-1$ items with $v_i(S) = -|S|$ for every i leaves some agent empty-handed under any complete allocation, and each of the $n-1$ loaded agents envies that agent by 1. Either way, if Conjecture 3 holds then its bound is tight.

1.5 General binary is not a doubly monotone class

Recall that an instance is *doubly monotone* when, for each agent i and each item g, either $v_i(g \mid S) \geq 0$ for every S (the item is a *good* for i) or $v_i(g \mid S) \leq 0$ for every S (a *chore* for i). Both boundary classes of Observation 4 are doubly monotone; their common generalization is not.

Lemma 10. *There is a general binary valuation that is not doubly monotone.*

Proof Let $M = \{a, b, g\}$ and define

$$v(S) := \begin{cases} |S| & \text{if } |S| \leq 2, \\ 1 & \text{if } S = M. \end{cases}$$

Then $v(\emptyset) = 0$, and every marginal lies in $\{-1, 0, 1\}$: adding an item to a set of size 0 or 1 raises the value by exactly 1, and adding the third item to a set of size 2 lowers it from 2 to 1. So v is general binary. But $v(g \mid \emptyset) = 1 > 0$ while $v(g \mid \{a, b\}) = 1 - 2 = -1 < 0$, so g is neither a good nor a chore for this agent. □

Remark 11. Lemma 10 has a concrete consequence for what may be assumed. The reduction of Kawase, Makino, Sumita, Tamura and Yokoo [KMS+24], which converts any EF1 allocation into an envy-free one and yields $n - 1$ per agent, is stated for doubly monotone valuations under a two-sided normalisation. It therefore applies to each boundary class of Observation 4 separately — and did supply the baseline against which the negative dichotomous result was measured — but it does *not* apply to general binary instances off the shelf.

2 Ideas Inbox

Empty. Append below as thinking starts.

References

[BKNS22] Siddharth Barman, Anand Krishna, Y. Narahari, and Soumyarup Sadhukhan. Achieving envy-freeness with limited subsidies under dichotomous valuations, 2022.

[HS19] Daniel Halpern and Nisarg Shah. Fair division with subsidy. In *Proceedings of the 12th International Symposium on Algorithmic Game Theory (SAGT)*, volume 11801 of *Lecture Notes in Computer Science*, pages 374–389. Springer, 2019.

[KMS+24] Yasushi Kawase, Kazuhisa Makino, Hanna Sumita, Akihisa Tamura, and Makoto Yokoo. Towards optimal subsidy bounds for envy-freeable allocations. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024.